USCG CAPTAIN'S LICENSE · OUPV NEAR COASTAL

Navigation General

Visual study guide — diagrams, memory hooks & exam traps



Compass & TVMDC · Charts · Tides & currents · Buoys & lights · Light ranges
Piloting math · GPS, radar & depth sounders · Weather · Right whale rules
Built from 101 MLS Nav General cards + 48 official USCG Q171 sample questions
Prepared for Jesse Orebaugh · October 2026

How this exam module works

Navigation General (Q171) tests the knowledge you need to know where you are and where the water is safe: the compass, charts, tides, buoys and lights, electronics and weather. The passing mark is **70%**. Unlike Deck General, a slice of it is **arithmetic**: compass conversions, light visibility, and course and ETA problems. Those are free points once the method is automatic.

- **Write TVMDC down** the moment you open a compass question, fill in what's given, then add or subtract (page 4 walks through it).
- **Label every number** with T, M or C and E or W. Most wrong answers come from applying a correction the wrong way.
- **Pick the closest answer** on math problems. The bank rounds (9.1 mi is keyed 9.0).
- **The chart plot is a separate module** (Q172, plotted on a real chart). This guide covers Q171 only.

The module at a glance

Topic	MLS cards	What gets tested most	Page
Magnetic compass	18	TVMDC, variation vs deviation, compass error, lubber's line	3–5
Charts & position	14	Mercator, 1' latitude = 1 nm, lettering, datum, compass rose	6
Tides & currents	19	Spring/neap, flood/ebb, set and drift, range, datum	7
Aids to navigation	33	Red right returning, special marks, light rhythms, ranges	8
Piloting math	Q171	Leeway, current vectors, passing abeam, light visibility	10
Electronic navigation	16	GPS/DGPS, depth sounders, radar, AIS, chart plotters	11
Weather & regulations	Q171	Barometer, hurricanes, warnings, fog, right whales	12

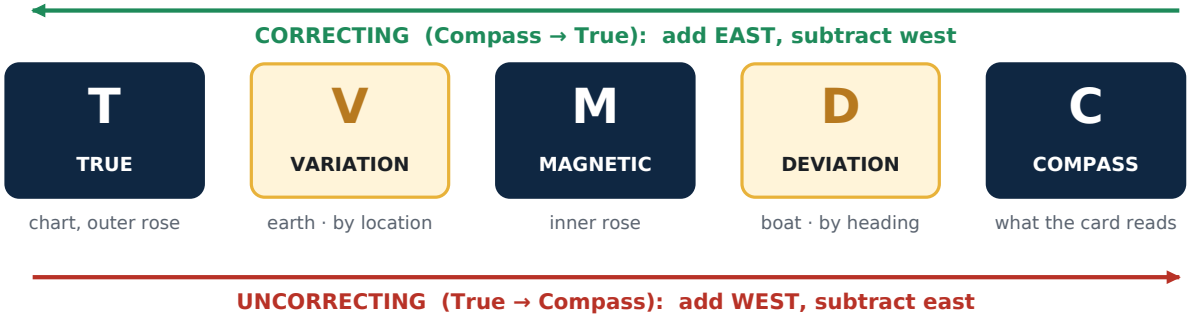
The 48 Q171 samples add the math, weather and AIS questions that MLS covers lightly.

TIP **How to use this guide with the app**

Read a page, then filter **Nav Gen** + that topic in Quiz mode, then finish with **USCG Samples** filtered to Q171. Do the worked problems on pages 4 and 10 by hand until you can do them without looking.

Magnetic compass I: TVMDC

A compass points to **magnetic** north, and the boat's own magnetism bends it further. Two corrections connect the compass to the chart:



True Virgins Make Dull Companions · Can Dead Men Vote Twice

Correction	Caused by	Changes with	Where you find it
Variation	The earth: true and magnetic north aren't in the same place	**Geographic location** (and slowly over years)	Center of the chart's compass rose
Deviation	The vessel: engine, steel, electronics, wiring	**Vessel heading**	The boat's deviation card
Compass error	Variation + deviation combined algebraically	Both	Compute it

MEMORY HOOK **Compass least, error east · Compass best, error west**

If the compass reads **less** than magnetic (or true), the error is **east**. If it reads **more**, the error is **west**.

Worked examples from the bank

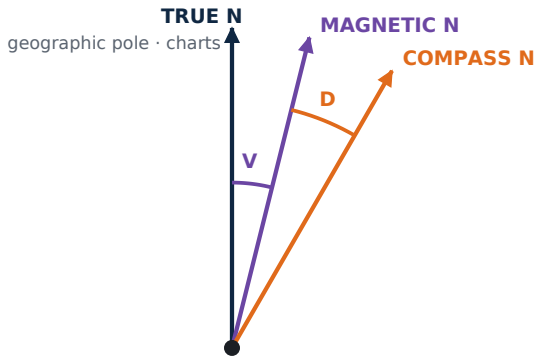
Problem	Work it	Answer
Range bears 229°T, variation 6°W, compass reads 232°. Deviation?	T 229 + 6W = M 235. Compass 232 is less than 235	**3°E**
Range bears 014°T, variation 11°E, compass reads 009°. Deviation?	T 014 − 11E = M 003. Compass 009 is more than 003	**6°W**
Compass and magnetic headings are the same	Nothing between them	**No deviation** on that heading

EXAM TRAP **Deviation follows HEADING, variation follows LOCATION**

A question says the correction changed when you changed course? That's **deviation**. Changed as you moved across the chart? That's **variation**.

TVMDC step by step

Every heading is an angle measured from some "north." TVMDC converts one direction between three different norths.



VARIATION = true ↔ magnetic
Caused by the earth. Changes as you MOVE.
Read it from the chart's compass rose.

DEVIATION = magnetic ↔ compass
Caused by the boat. Changes as you TURN.
Read it from the deviation card.

Here both pull the needle EAST of true,
so the compass reads a SMALLER number.

MEMORY HOOK **You already know this from flying**

Pilots go true course → magnetic heading with **"East is least, West is best."** On the boat that same rule is the whole trip from the chart down to the compass, applied twice: once for variation, once for deviation.

Why east is subtracted going toward the compass

An **easterly** error means the needle's north has been pulled to the east. Every heading is then measured from a north that's already partway around the dial, so the compass shows a **smaller** number. That's where both rules come from:

- **Chart → compass (uncorrecting):** east is least, west is best. Subtract east, add west.
- **Compass → chart (correcting):** the reverse. Add east, subtract west.
- **Finding an unknown error:** compass least, error east. Compass best (bigger), error west.

Write the ladder every time

Chart → compass (uncorrecting)		
T	200°	plotted on the chart
V	5°W	going down: west is best, ADD
M	205°	
D	2°E	going down: east is least, SUBTRACT
C	203°	← steer this

Compass → chart (correcting)		
C	090°	read off the compass
D	3°W	going up: SUBTRACT west
M	087°	
V	10°E	going up: ADD east
T	097°	← plot this

The bank's favorite: find deviation on a range

When a range is in line, you know the boat's true heading exactly, so the only unknown is deviation.

Solve for deviation with a range		
T	229°	range bearing from the chart
V	6°W	add west going down
M	235°	
D	?	compass 232 is LESS than 235 → error EAST = 3°E
C	232°	what the compass read on the range

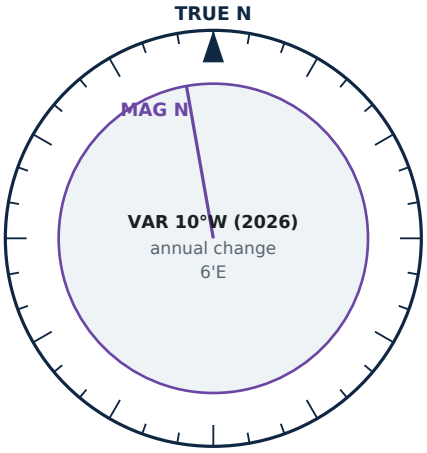
TIP **Three habits that prevent wrong answers**

1. Write T-V-M-D-C down the side of the scratch paper before reading the choices.

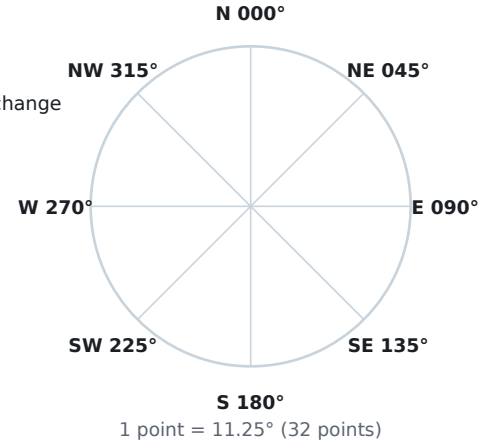
2. Fill in what's given, then go **one step at a time**. Never apply both corrections at once.

3. Sanity check: the answer differs from the start by only a few degrees, in the direction the errors push.

Magnetic compass II: the rose and the card



outer ring = TRUE
inner ring = MAGNETIC
center = variation + yearly change



1 point = 11.25° (32 points)

Updating variation for the year

The rose prints variation for a base year plus an annual change. Multiply years by the change and apply it.

Bank problem	Work it	Answer
3°45'E in 1988, increasing 6' per year. Use in 1991?	3 years × 6' = 18' more. 3°45' + 18'	**4°03'E**

Compass facts

Question	Keyed answer
Operating principle	**Unlike poles attract**
Card marked in	**360°**, clockwise from north
Lubber's line shows	**The direction the bow is heading** (fixed in line with the keel)
Turning to starboard, the card	**Stays aligned with compass north**; the boat turns around it
Primary advantage	**Reliable due to its simplicity** (no power needed)
Brass gimbals and binnacles	Brass **cannot be magnetized**
Things that affect it	Iron, electronics, current-carrying wires: **all of these**
Unused for a long time	**Shield it from direct sunlight**
Hand-bearing compass deviation	Taken as **0°** (apply variation only)
Bearings can be expressed as	**Compass, magnetic or true**

Charts and position

The Mercator chart

- Mercator is a **cylindrical** projection, the only cylindrical one widely used for navigation. Rhumb lines are straight.
- **1 nautical mile = 1 minute of latitude.** Measure distance on the **latitude scale** (side of the chart), **near the middle of the track**, because the scale stretches with latitude. Never use the longitude scale.
- A **knot** is one nautical mile per hour (never "knots per hour").

Reading the chart

Question	Keyed answer
Slanted (italic) lettering	Feature is **underwater or floating**
Vertical (upright) lettering	Feature is **above water**: a rock named in vertical letters is high and dry
Conspicuous landmarks	Labeled in **CAPITAL letters**
Found in the chart title	Projection, scale and the **sounding datum**
Compass rose outer ring	**True** directions
Chart reprinted with extensive corrections	**New edition** (cancels the old one)

Depths and datum

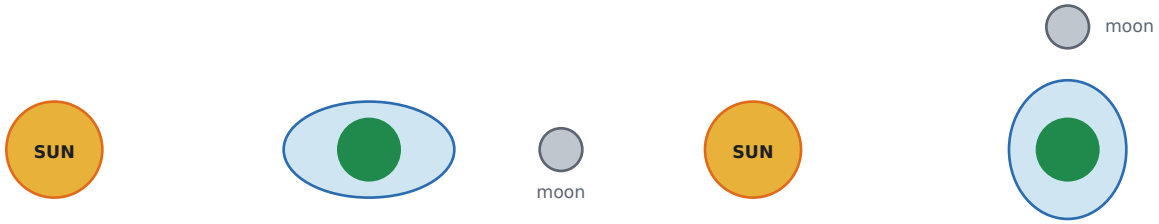
Term	Meaning
Chart (sounding) datum	The baseline for **every sounding** on the chart
U.S. datum (East Coast and all U.S. coasts)	**Mean lower low water (MLLW)**
Charted depth	Vertical distance from the **datum to the bottom**
Height of tide	Distance from the **datum to the water surface**
Actual depth	Charted depth + height of tide

MEMORY HOOK **Slanted letters are swimming, standing letters are standing**

Italic type leans over like it's sinking: submerged or floating. Upright type stands on dry ground.

Tides and currents

Tide is the vertical rise and fall. **Current** is horizontal flow. Tidal currents go with the rise and fall of the tide.



SPRING TIDES
new & full moon · sun, earth, moon in line · biggest range

NEAP TIDES
quarter moons · at right angles · smallest range

Term	Meaning
Spring tide	Sun and moon in line: **highest highs, lowest lows** (biggest range)
Neap tide	Sun and moon at right angles: **lower highs, higher lows**
Semidiurnal	**Two** highs and two lows each tidal day: the U.S. East Coast
Diurnal	**One** high and one low each tidal day
Range of tide	Difference between the heights of high and low water
Flood current	Flows **toward land** as the tide rises
Ebb current	Flows **away from land** as the tide falls
Set	**Direction** the current flows toward
Drift	**Speed** of the current in knots
Rotary current (off the East Coast)	Turns through every direction once per tidal cycle: about **12½ hours**

MEMORY HOOK **Flood floods in, Ebb exits · Set steers, Drift drives**

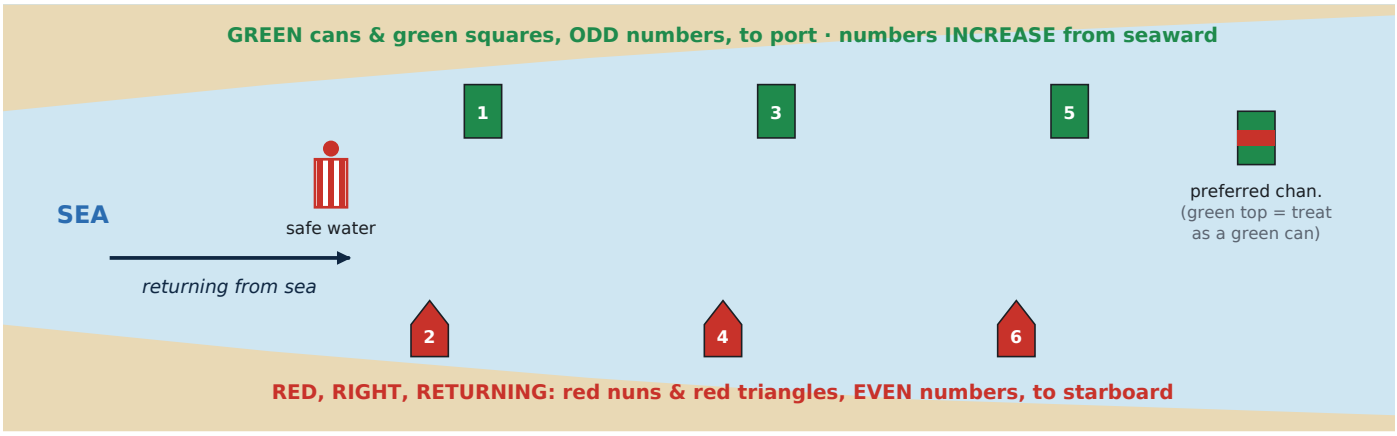
Set is a direction (where it's going), drift is a speed.

Using the tables

Question	Keyed answer
Time of high tide at a place and date	**Tide Tables** (Current Tables give flow, not heights)
Aground in soft mud: best chance to refloat if you grounded at	**Low water springs** (every later tide is higher)
Going with the current	Speed over ground is **faster** than speed through the water

Aids to navigation I: buoys and daymarks

The U.S. uses **IALA Region B**: red to starboard returning from sea.



Mark	Color / shape	Numbers	Light	Daymark
Starboard (right returning)	Red **nun** (cone top)	**Even**	Red	Red **triangle**
Port (left returning)	Green **can**	**Odd**	Green	Green **square**
Preferred channel (junction)	Red and green horizontal bands	Letters	Matches the top band	Banded
Safe water (mid-channel)	Red and white vertical stripes	Letters	White Mo(A)	Red-white **octagon**

EXAM TRAP **Preferred channel: read the TOP band**

The top band tells you which lateral mark it acts like. Green top = treat it like a green can (keep it to port returning). Its reflective tape and light match the top band too.

Non-lateral marks



Safe water
red/white vertical
white Mo(A) or Iso
red sphere topmark



Isolated danger
black/red bands
white Fl(2)
two black spheres



Special mark
yellow
yellow light
dredging, cables, anchorage



Mooring buoy
white, blue band
white light
-

Regulatory marks (white with orange)



DANGER
Diamond
rock, wreck, dam



EXCLUSION
Crossed diamond
boats keep out



CONTROLLED AREA
Circle
speed limit, no wake



INFORMATION
Rectangle
directions, distances

Aids to navigation II: lights and ranges

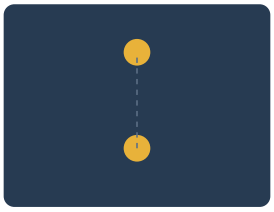
Light rhythms

Fixed (F)	steady, never off
Flashing (Fl)	light SHORTER than dark
Occulting (Oc)	light LONGER than dark
Isophase (Iso)	light = dark (e.g. 1 s on, 1 s off)
Quick (Q)	about 60 flashes per minute
Long flash (L Fl)	flash of at least 2 seconds
Group Fl(2)	isolated danger (white)
Morse A Mo(A)	short-long: safe water (white)

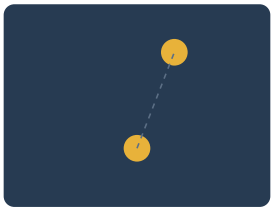


- **Period** = time for one complete cycle (Fl 4s means the pattern repeats every 4 seconds).
- Standard aid light colors: **red, green, white and yellow**. Blue isn't one.
- A **red sector** marks a **danger area** (shoals or land). Sector limits are **true bearings from the ship toward the light**.
- The **Light List** is published by the **U.S. Coast Guard** and includes authorized private aids. (NGA publishes the foreign List of Lights.)

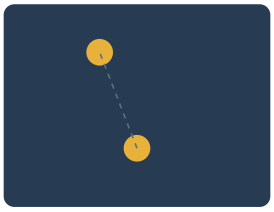
Range lights



ON the range
lights in line: hold course



Top light to the RIGHT
steer right (toward the top light)



Top light to the LEFT
steer left

When will I see the light?

Range	Meaning
Nominal range	How far the light shows in clear weather (10 nm visibility), from the Light List
Luminous range	Nominal range adjusted for the **existing visibility**
Geographic range	Limited by the earth's curve: **1.17 × (√height of light + √height of eye)**, feet in, nm out
You first sight it at	The **lesser** of luminous and geographic range

TIP **Bank problem: 10 nm nominal light, 11 ft high, visibility 15 nm, height of eye 20 ft**

Geographic = 1.17 × (3.32 + 4.47) = **9.1 nm**. Luminous range with 15 nm visibility is more than 10 nm, so the geographic range limits you: **about 9.0 miles**.

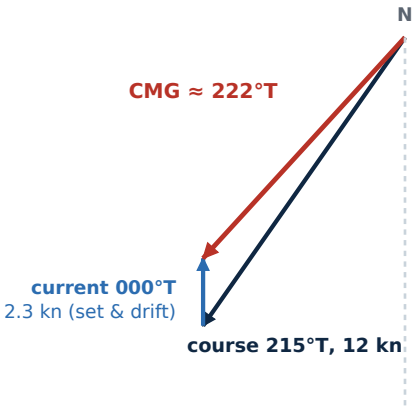
Piloting math

Leeway: wind pushes you downwind

Convert compass to true first (add east, subtract west), then apply leeway **away from the wind**. Wind on the port side pushes you to starboard: **add**. Wind on the starboard side: **subtract**.

Bank problem	Work it	Answer
149° psc, var 13°E, dev 4°E, wind NE, 4° leeway	C 149 + 4E = M 153 + 13E = T 166. NE wind hits the port side: +4	**170°T**
299° psc, var 7°W, dev 4°W, wind SW, 3° leeway	C 299 − 4W = M 295 − 7W = T 288. SW wind hits the port side: +3	**291°T**

Current: course made good

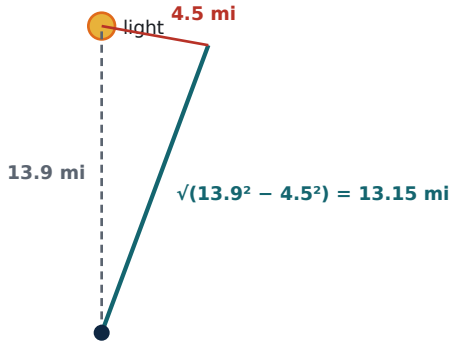


Steer the boat vector, add the current vector tip-to-tail. The line from start to the end is course and speed made good.

Set = direction the current flows TOWARD
Drift = its speed in knots

A current from astern raises speed over ground; a head current lowers it.

Passing a light abeam



Right triangle: hypotenuse = range to light, short side = passing distance.

Distance to abeam = √(range² − off²)
Course offset = arcsin(off ÷ range)

13.15 mi ÷ 21 kn = 37.6 min
0054 + 38 min = ETA 0131

Leave the light to starboard → steer LEFT of its bearing by the offset angle.

Bank problem	Work it	Answer
Course 321°T, light 7° on the starboard bow at 9.7 mi. Course to pass 3.5 mi abeam, light to starboard?	Bearing 328°T. Offset = arcsin(3.5 ÷ 9.7) = 21°. Leave it to starboard: 328 − 21	**307°T**
Light dead ahead 13.9 mi at 0054, pass 4.5 mi off at 21 kn. ETA abeam?	√(13.9² − 4.5²) = 13.15 mi ÷ 21 = 37.6 min	**0131**

Electronic navigation

GPS

Question	Keyed answer
What GPS is	Satellite radio navigation **available 24 hours a day, worldwide, in all weather**
Expected GPS accuracy	Within about **20 meters** (bank key; modern units do better)
System accurate to 3 m with 99.7% probability	**DGPS** (bank key)
USCG Maritime DGPS accuracy in coverage	**10 meters** (bank key; service ended in 2022)
Tool that has largely replaced paper charts	**Electronic chart plotter**
A chart plotter shows	Position, track history and heading: **all of these**
Heading	The direction the vessel is **pointing at any given moment**

Depth sounders

Question	Keyed answer
Sound pulses are produced by a	**Transducer**
Operating principle	Sound travels at a **constant speed** through water; depth = speed × time ÷ 2
Depth measured from	**The bottom of the transducer** to the seabed
Offset function	Adjusts the display to read from the waterline or below the keel
Over a muddy bottom	May show **two depth readings** (top of the mud and hard bottom)

Radar

Question	Keyed answer
NOT a limitation	It **can be used in limited visibility** (that's its strength)
FALSE statement	Detection is **unaffected by weather** (rain and sea clutter mask targets)
Best echo	A **300-ft freighter, beam on**
In fog	**Fiberglass and small wooden boats** may not show up

AIS

Question	Keyed answer
Short text message length	**158 characters**
What it automatically provides	Identity, type, position, course, speed, status and voyage data: **all of the above**

Weather and regulations

Barometer and pressure

Question	Keyed answer
Numbers on isobars	**Barometric pressure**
Reports give pressure at	**Sea level**
Aneroid barometer units	**Inches of mercury and millibars**
High approaching	Barometer **rises**
Rapid rise	**Clearing weather, possibly with high winds**

Clouds, fog and warnings

Question	Keyed answer
"Nimbo" means	**Rain** clouds
Classic thunderhead	**Cumulonimbus**
Clear sky with a few cumulus	**Fair weather**
Advection fog	**Warm moist air blown over a colder surface**
Fog clears when	Temperature rises, wind increases, or drier air arrives: **all of the above**
Gale, storm and hurricane warnings are based on	**Forecast wind speed** (gale 34–47 kn, storm 48–63, hurricane 64+)

Hurricanes

Question	Keyed answer
FIRST sign of a well-developed tropical cyclone	An **unusually long ocean swell**
Dense black clouds around the eye	**Wall clouds**
Average speed after recurvature	**20 to 30 knots**

North Atlantic right whales

> **REGULATION** **500 yards**
>
> It's unlawful to approach within **500 yards**. If you find yourself inside 500 yards, **turn away and leave at a slow, safe speed**.

Notices

A **HYDROLANT** warns of hazards on the high seas, not local harbor matters. An extinguished light inside a harbor (the bank uses Robbins Reef Light in New York's Upper Bay) goes out by Local Notice to Mariners, so it's the EXCEPT answer.

Your trouble spots

The 4 Nav General questions you missed on MLS

Topic	The fact to lock in
Compass	Unused for a long time: **shield it from direct sunlight** (never remove the correctors)
Compass	Lubber's line = **the direction the bow is heading**
Compass	Turning to starboard, the card **stays aligned with compass north**
GPS	Expected accuracy: within **20 meters**

Bank keys: memorize the expected answer

Question	Bank answer	Why it's debatable
GPS accuracy radius	**20 meters**	Modern GPS is usually under 10 m
3 m with 99.7% probability	**DGPS**	Official spec was ~10 m (95%); service ended 2022
USCG Maritime DGPS accuracy (Q171)	**10 meters**	Correct spec, but the service is discontinued
What largely replaced paper charts	**Chart plotter**	Paper is still a prudent backup

EXAM TRAP **Two DGPS questions, two numbers**

If the stem says "3 meters with 99.7% probability," the answer is **DGPS**. If it asks the accuracy of USCG Maritime DGPS, the answer is **10 meters**.

Quick-recall sheet

Fact	Value
TVMDC correcting (C → T)	Add east, subtract west
NE / SE / SW / NW	045° / 135° / 225° / 315°
1 minute of latitude	1 nautical mile
U.S. chart datum	MLLW
Geographic range	1.17 × (√HoL + √HoE)
Long flash	At least 2 seconds
Safe water light / isolated danger light	White Mo(A) / white Fl(2)
Right whale	500 yards

Self-test (answers on next page)

1. Which correction changes with the vessel's heading, and which with location?

2. Compass reads 090°, deviation 3°W, variation 10°E. What's the true heading?

3. True course 200°, variation 5°W, deviation 2°E. What's the compass course?

4. A range bears 090°T, variation 4°W, and the compass reads 096°. What's the deviation?

5. The chart shows variation 2°30'W in 2020, increasing 4' a year. What do you apply in 2026?

6. What does the lubber's line show?

7. Where on a Mercator chart do you measure distance?

8. A charted rock is named in slanted letters. What does that tell you?

9. What's the datum for soundings on U.S. charts?

10. What causes spring tides, and how do they differ from neap tides?

11. Define set and drift.

12. Returning from sea, which side are the even-numbered buoys on, and what shape are they?

13. A buoy is banded red and green with green on top. How do you treat it?

14. Which mark shows a white Morse A light? Which shows white Fl(2)?

15. A white buoy has an orange diamond with a cross inside. What does it mean?

16. What's the difference between flashing and occulting?

17. On a range, the top light is drifting to your left. Which way do you steer?

18. Light 16 ft high, your eye 9 ft. Geographic range?

19. Course 300° psc, dev 2°E, var 6°W, wind from the north, 3° leeway. True course made good?

20. A light dead ahead is 10 mi away. You want to pass it 6 mi abeam. How far until it's abeam?

21. What produces the sound pulse in a depth sounder, and where is depth measured from?

22. Name one radar limitation in fog.

23. What's the FIRST sign of a well-developed tropical cyclone?

24. The barometer rises rapidly. What weather do you expect?

25. How close may you approach a North Atlantic right whale?

Answer key

1. **Deviation** changes with heading; **variation** with location.

2. C 090 − 3W = M 087 + 10E = **097°T**.

3. T 200 + 5W = M 205 − 2E = **203° compass**.

4. T 090 + 4W = M 094. Compass 096 is more (compass best): **2°W**.

5. 6 years × 4' = 24' more: **2°54'W**.

6. The direction the **bow is heading**.

7. On the **latitude scale**, near the middle of the track.

8. It's **underwater** (or floating).

9. **Mean lower low water** (MLLW).

10. Sun and moon **in line** (new and full moon): higher highs and lower lows. Neap (at right angles) has the smallest range.

11. **Set** = direction the current flows toward. **Drift** = its speed.

12. **Starboard**; red **nuns**.

13. Like a **green can**: keep it to port when returning from sea.

14. **Safe water** = Mo(A). **Isolated danger** = Fl(2).

15. **Exclusion area**: boats keep out.

16. Flashing: light **shorter** than dark. Occulting: light **longer** than dark.

17. Steer **left** (toward the top light).

18. 1.17 × (4 + 3) = **8.2 nm**.

19. C 300 + 2E = M 302 − 6W = T 296. North wind is on the starboard side heading 296, so you're pushed to port (−3): **293°T**.

20. $\sqrt{(10^2 - 6^2)}$ = **8 mi**.

21. A **transducer**; depth is measured from the bottom of the transducer.

22. Rain and sea clutter mask targets; **fiberglass and wooden boats** may not show up.

23. An **unusually long ocean swell**.

24. **Clearing**, possibly with high winds.

25. No closer than **500 yards**.

> **TIP** **Scoring yourself**
> Aim for **23/25**. If you miss any math question (2–5, 18–20), redo pages 4 and 10 before moving on.